



COMPUTER NETWORKS

TEAM NETWORKS

Department of Computer Science and Engineering

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Transport Layer

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Transport Layer

3.4 principles of reliable data transfer

3.5 connection-oriented transport: TCP

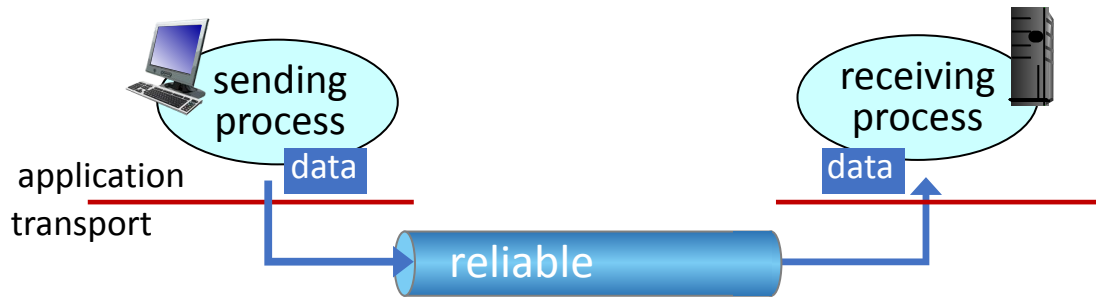
- segment structure
- reliable data transfer
- flow control
- connection management

3.6 principles of congestion control

3.7 TCP congestion control

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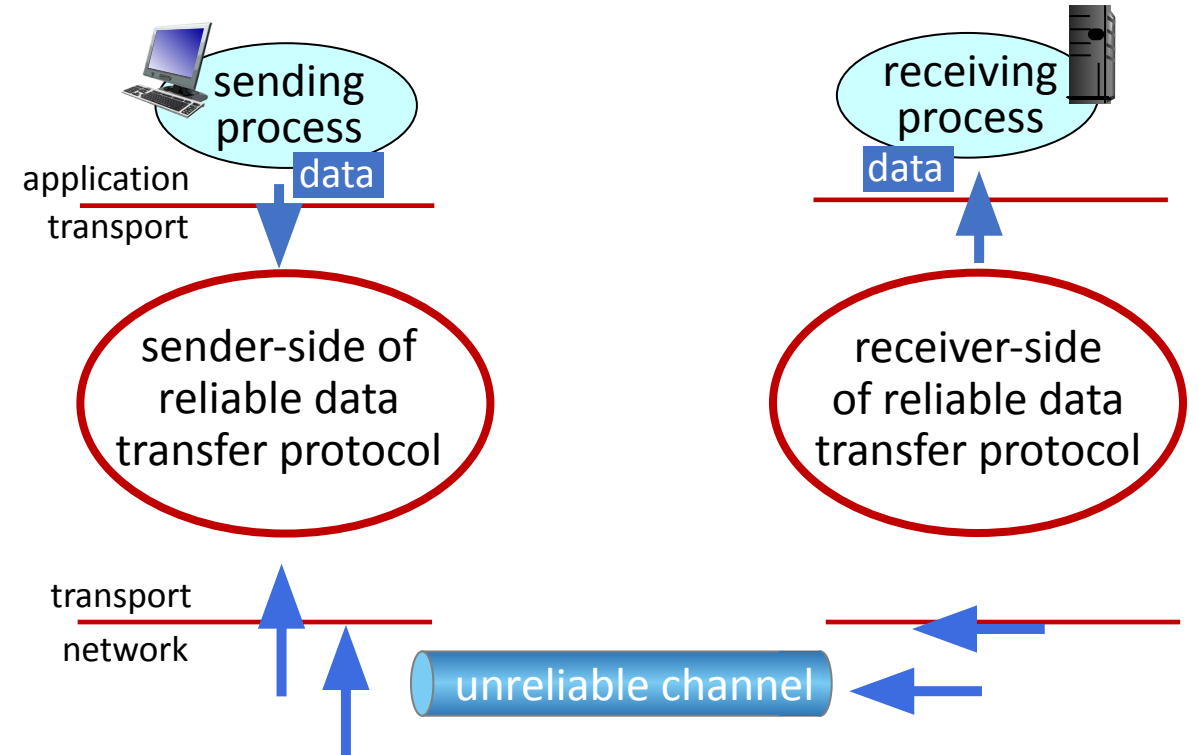
Principles of reliable data transfer



reliable service *abstraction*

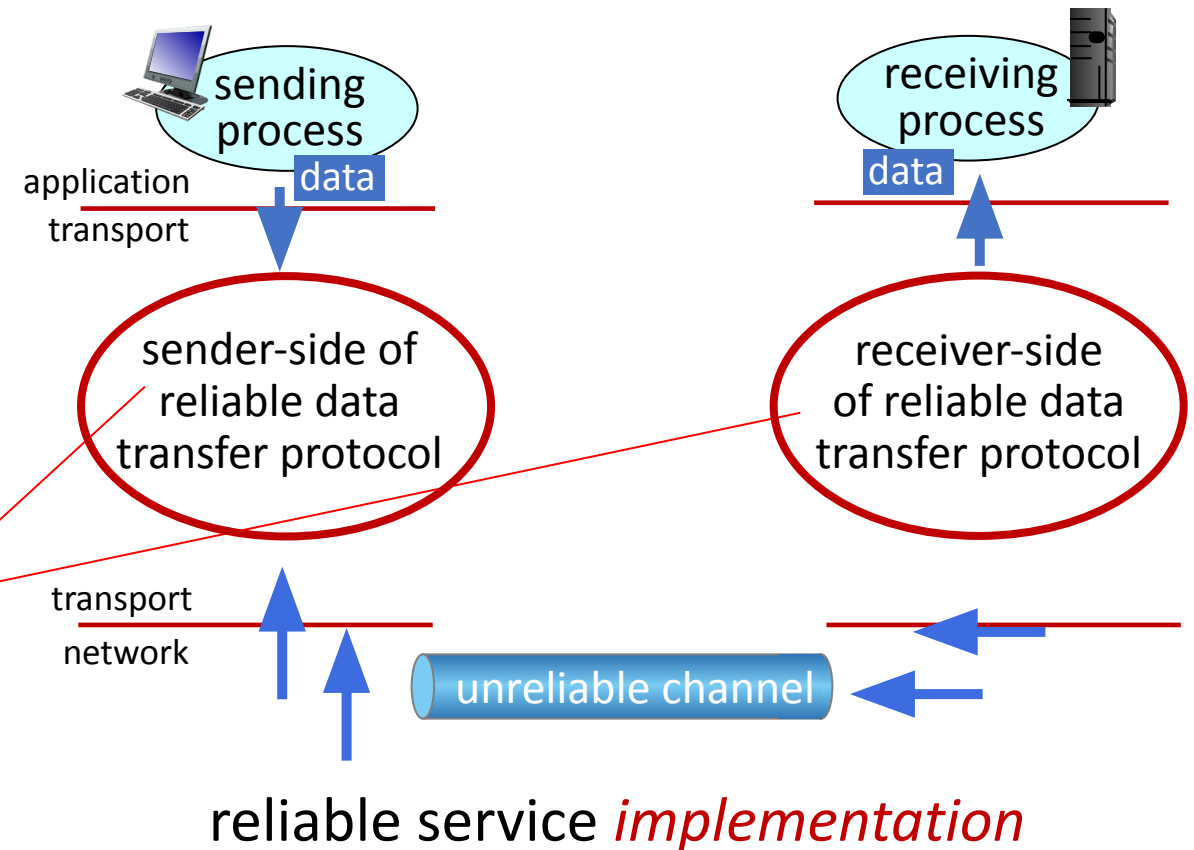


reliable service *abstraction*



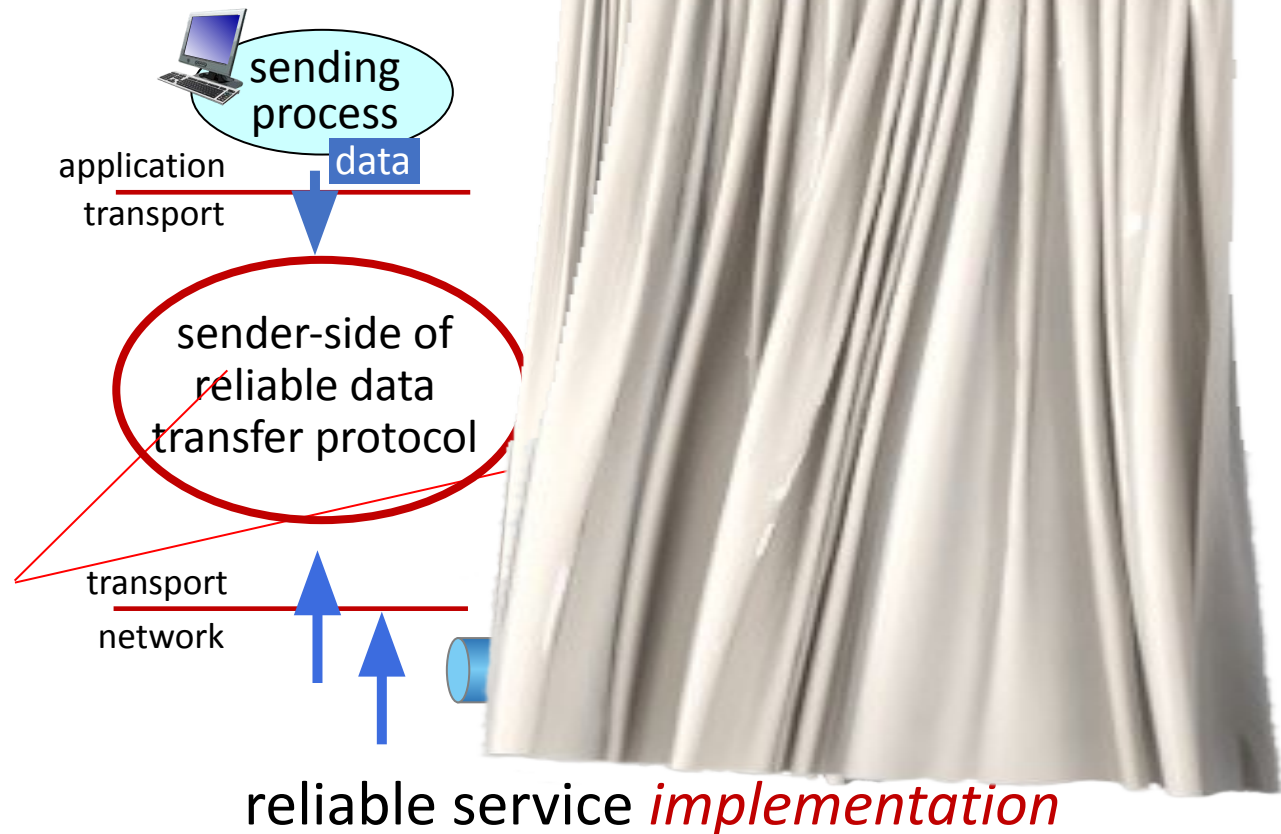
reliable service *implementation*

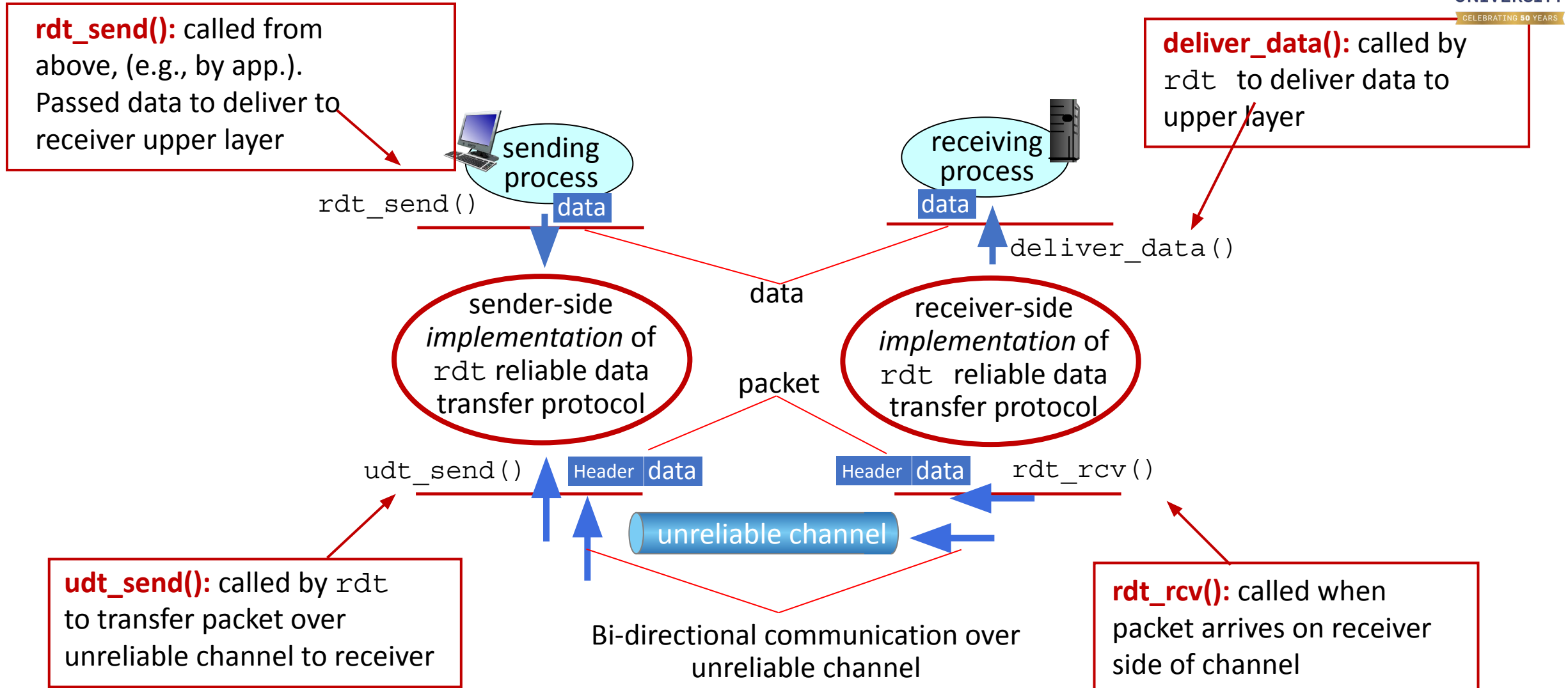
Complexity of reliable data transfer protocol will depend (strongly) on characteristics of unreliable channel (lose, corrupt, reorder data?)



Sender, receiver do *not* know the “state” of each other, e.g., was a message received?

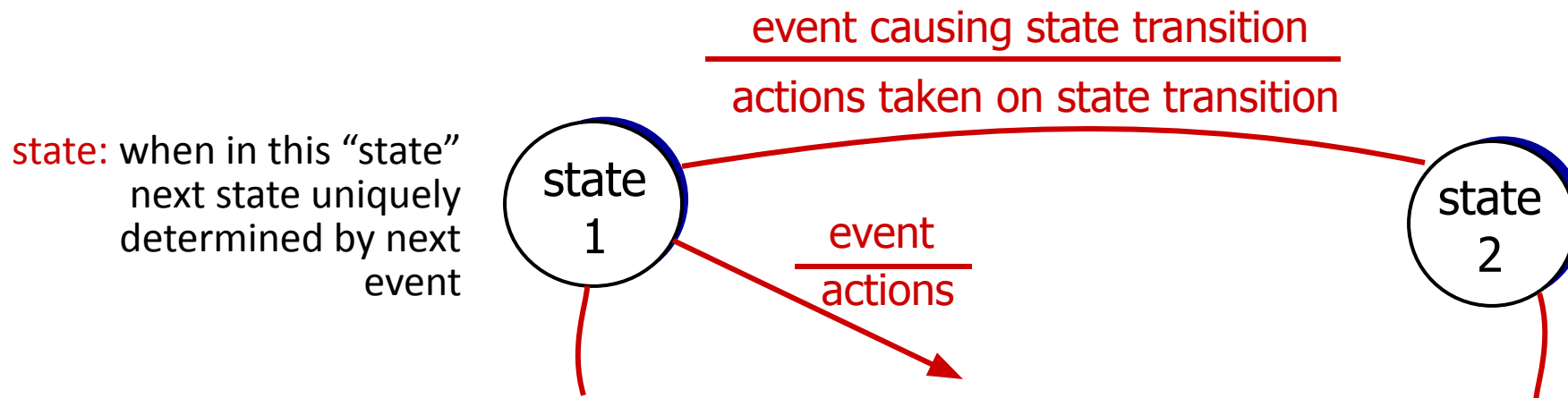
- unless communicated via a message



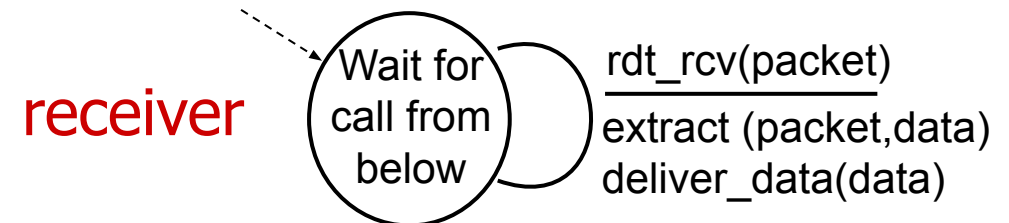
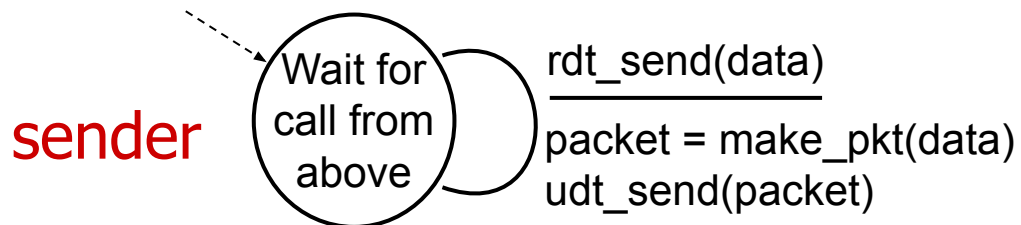


We will:

- incrementally develop sender, receiver sides of reliable data transfer protocol (rdt)
- consider only unidirectional data transfer
 - but control info will flow in both directions!
- use finite state machines (FSM) to specify sender, receiver



- underlying channel perfectly reliable
 - no bit errors
 - no loss of packets
- *separate* FSMs for sender, receiver:
 - sender sends data into underlying channel
 - receiver reads data from underlying channel



- underlying channel may flip bits in packet
 - checksum (e.g., Internet checksum) to detect bit errors
- *the* question: how to recover from errors?

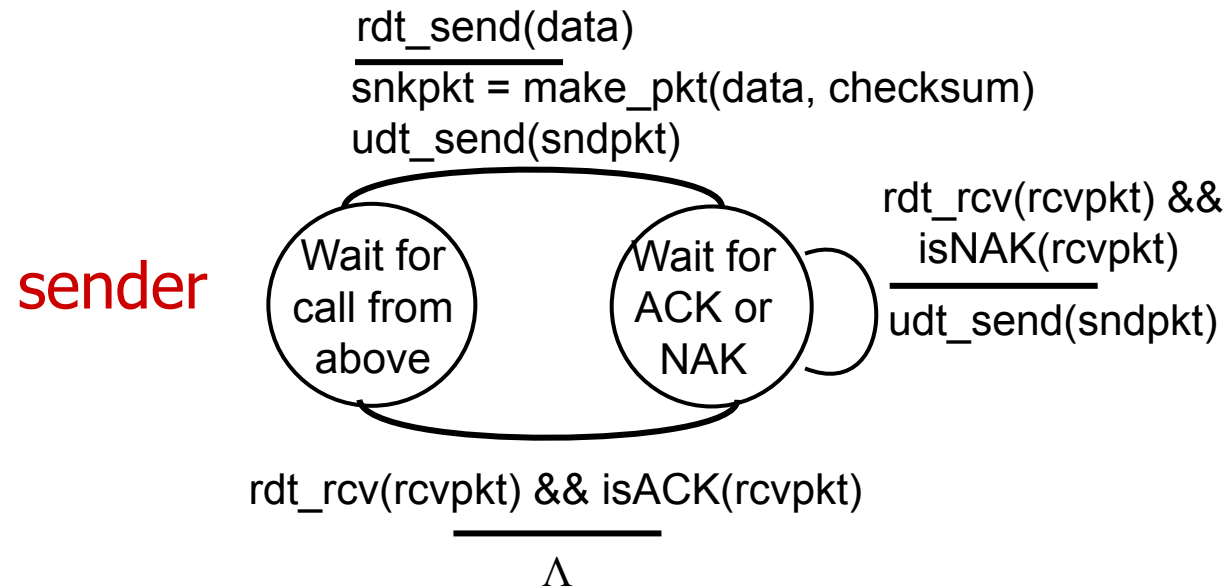
How do humans recover from “errors” during conversation?

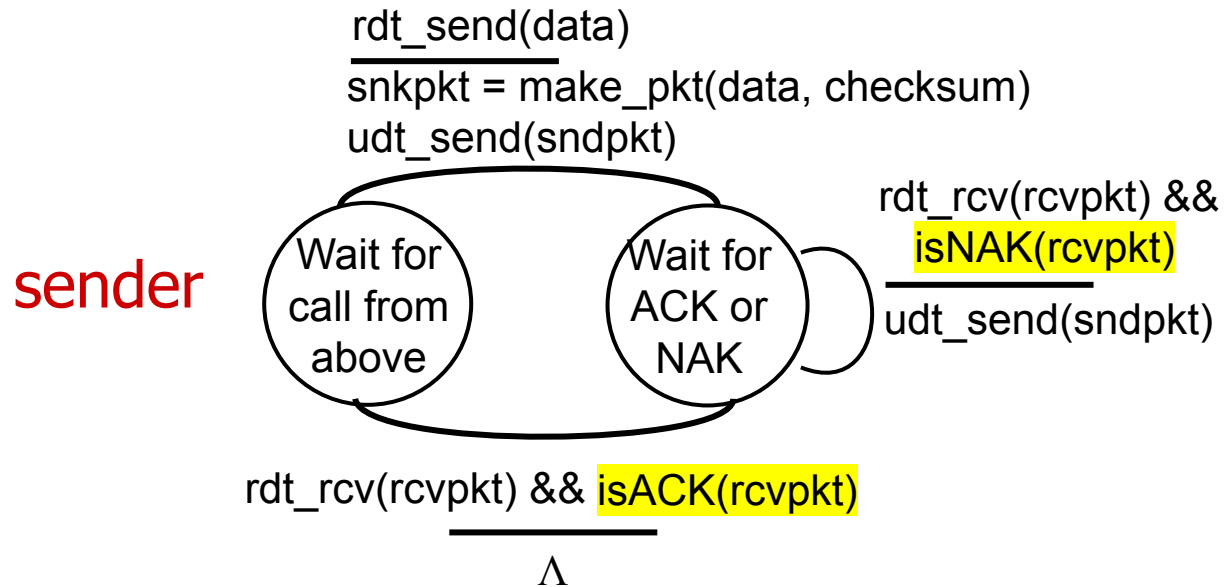
- underlying channel may flip bits in packet
 - checksum to detect bit errors
- *the question: how to recover from errors?*
 - *acknowledgements (ACKs)*: receiver explicitly tells sender that pkt received OK
 - *negative acknowledgements (NAKs)*: receiver explicitly tells sender that pkt had errors
 - sender *retransmits* pkt on receipt of NAK

**Automatic Repeat
reQuest**

stop and wait

sender sends one packet, then waits for receiver response



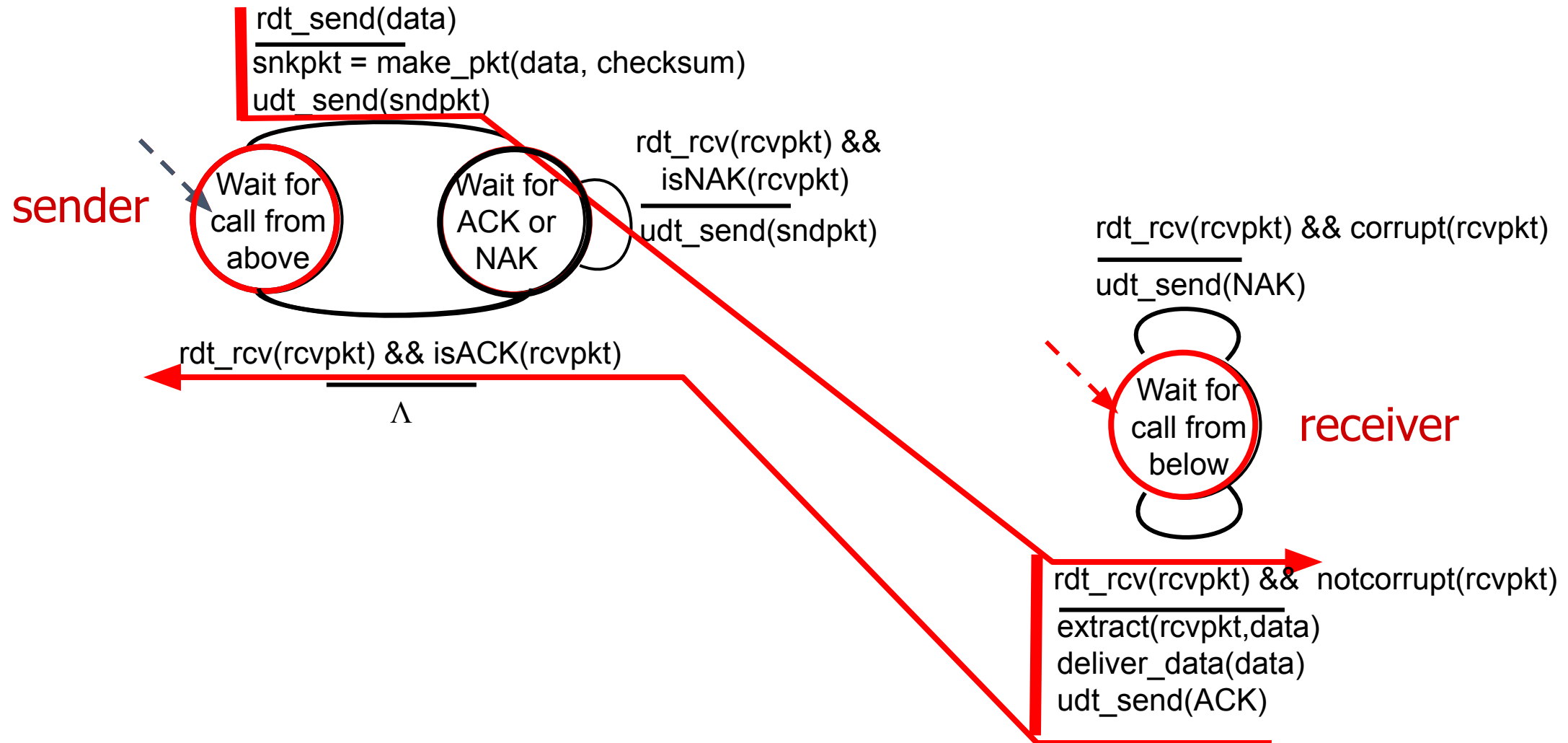


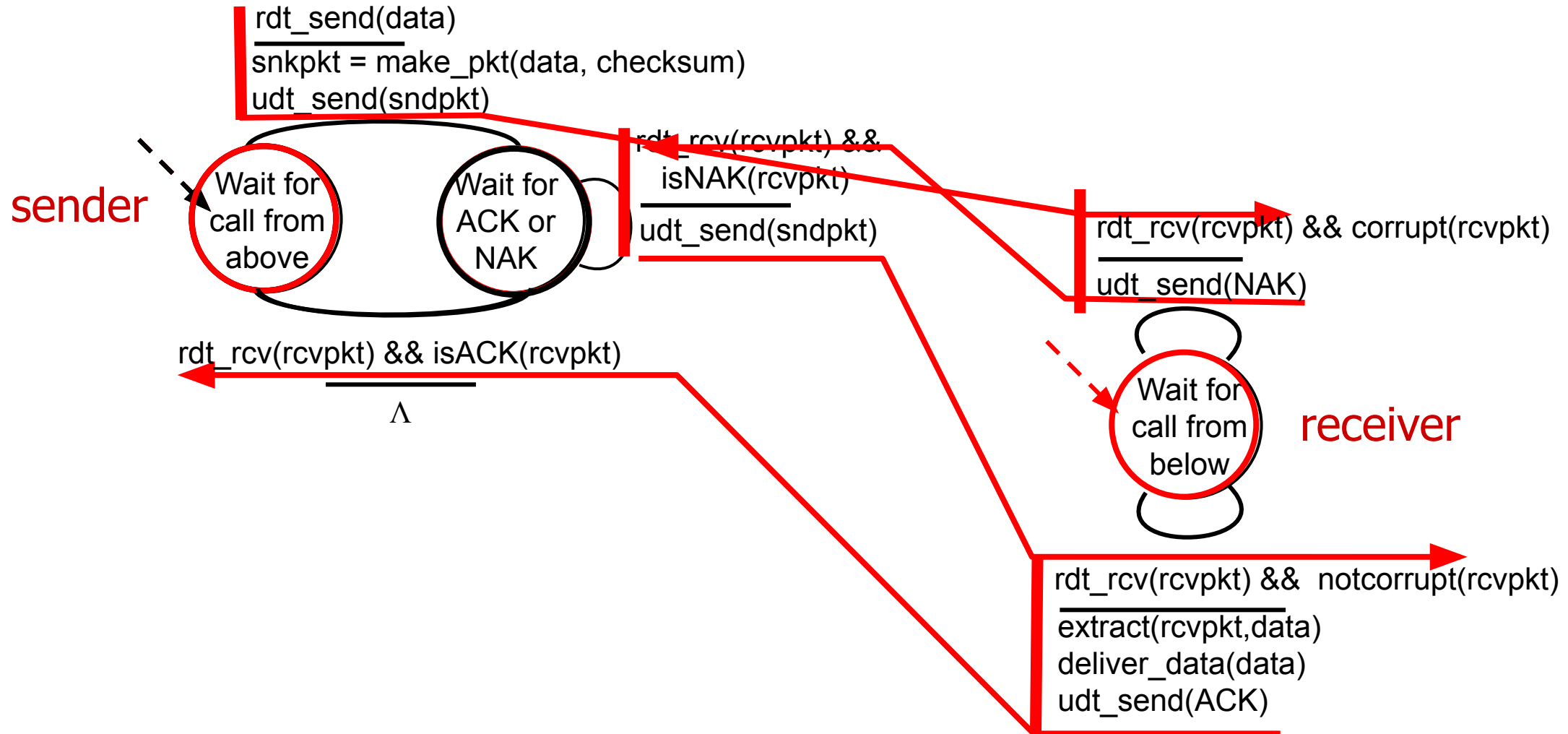
stop-and-wait

Note: “state” of receiver (did the receiver get my message correctly?) isn’t known to sender unless somehow communicated from receiver to sender

- that’s why we need a protocol!







what happens if ACK/NAK corrupted?

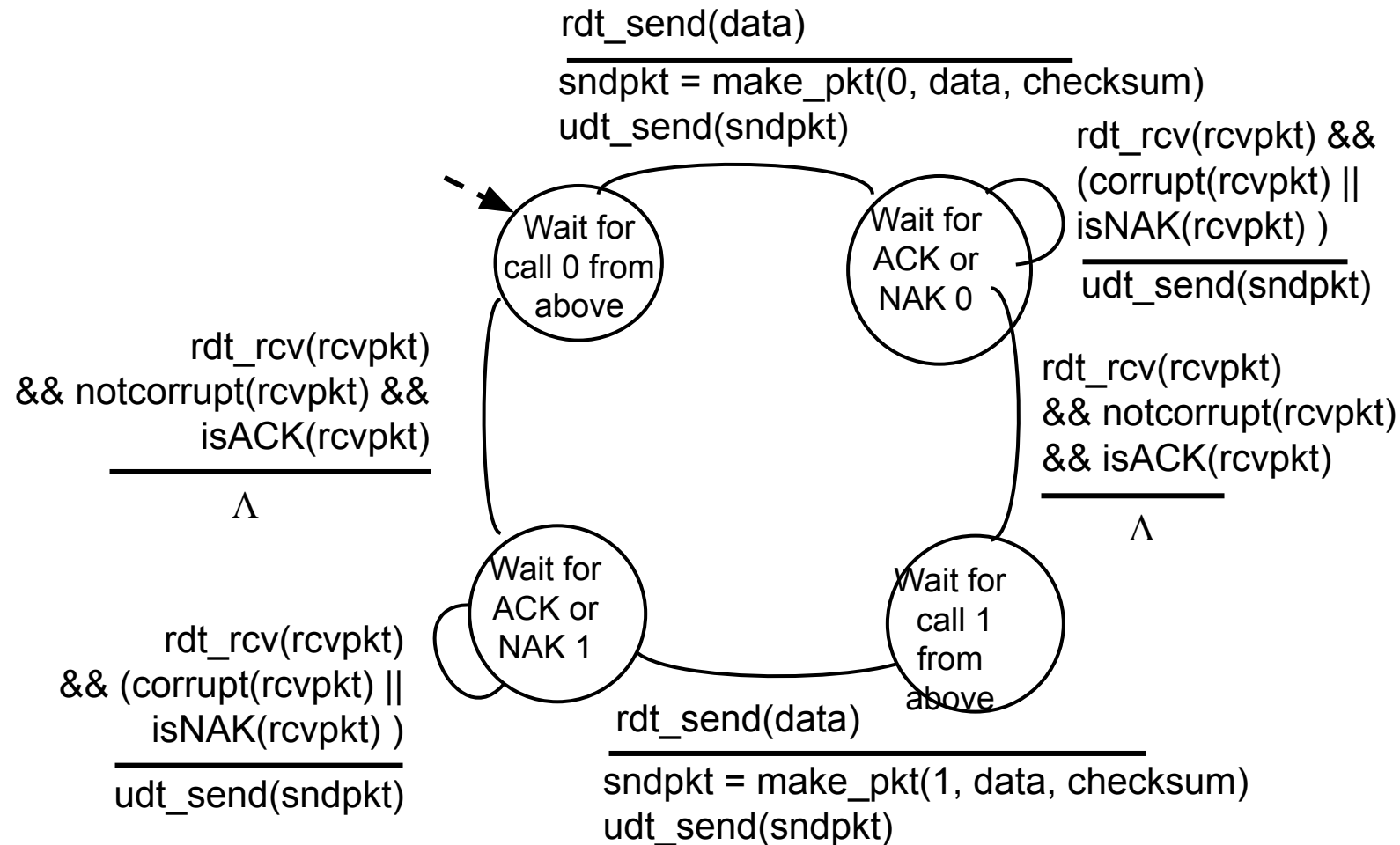
- sender doesn't know what happened at receiver!
- can't just retransmit: possible duplicate

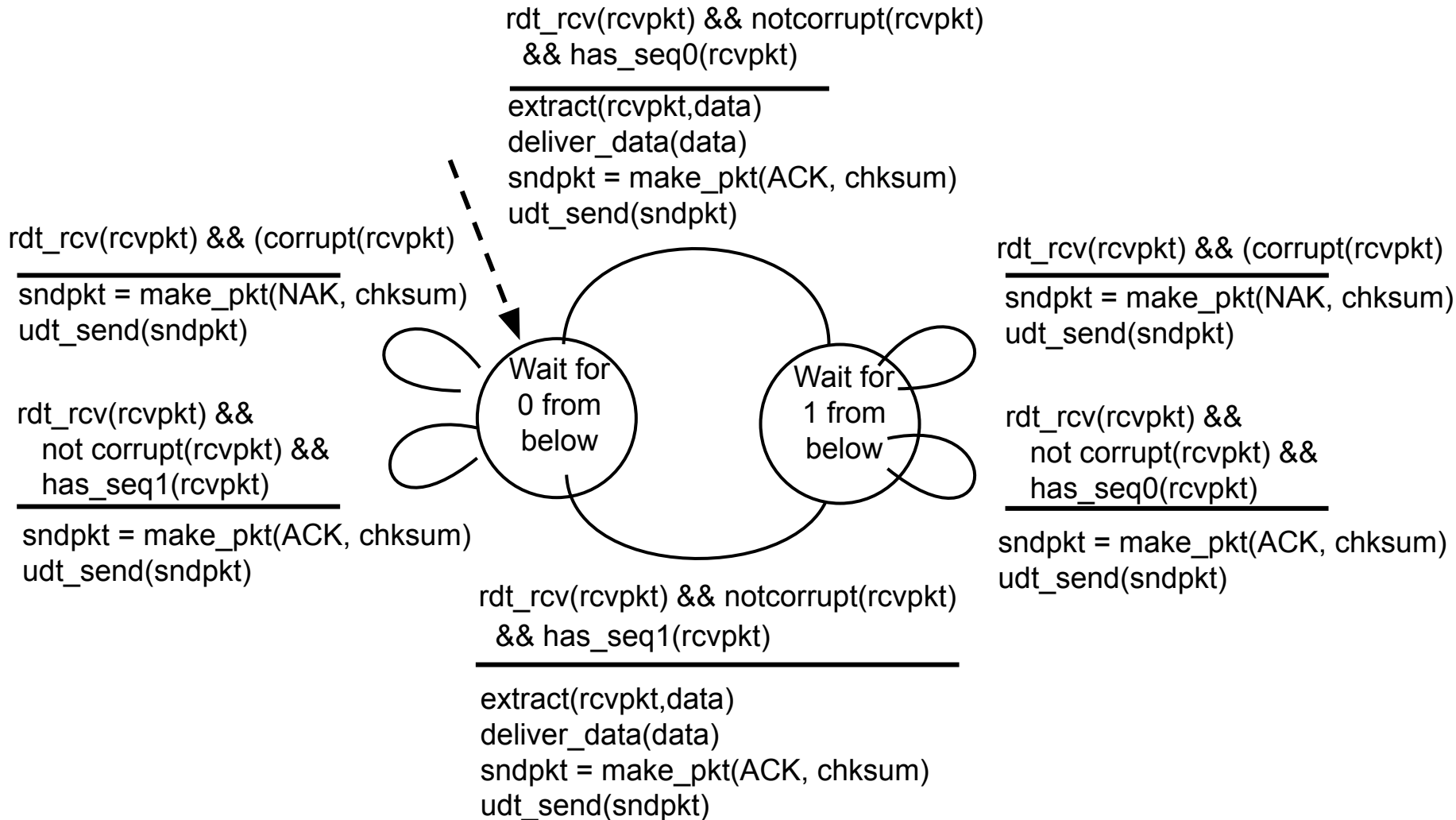
handling duplicates:

- sender retransmits current pkt if ACK/NAK corrupted
- sender adds *sequence number* to each pkt
- receiver discards (doesn't deliver up) duplicate pkt

stop and wait

sender sends one packet, then waits for receiver response





sender:

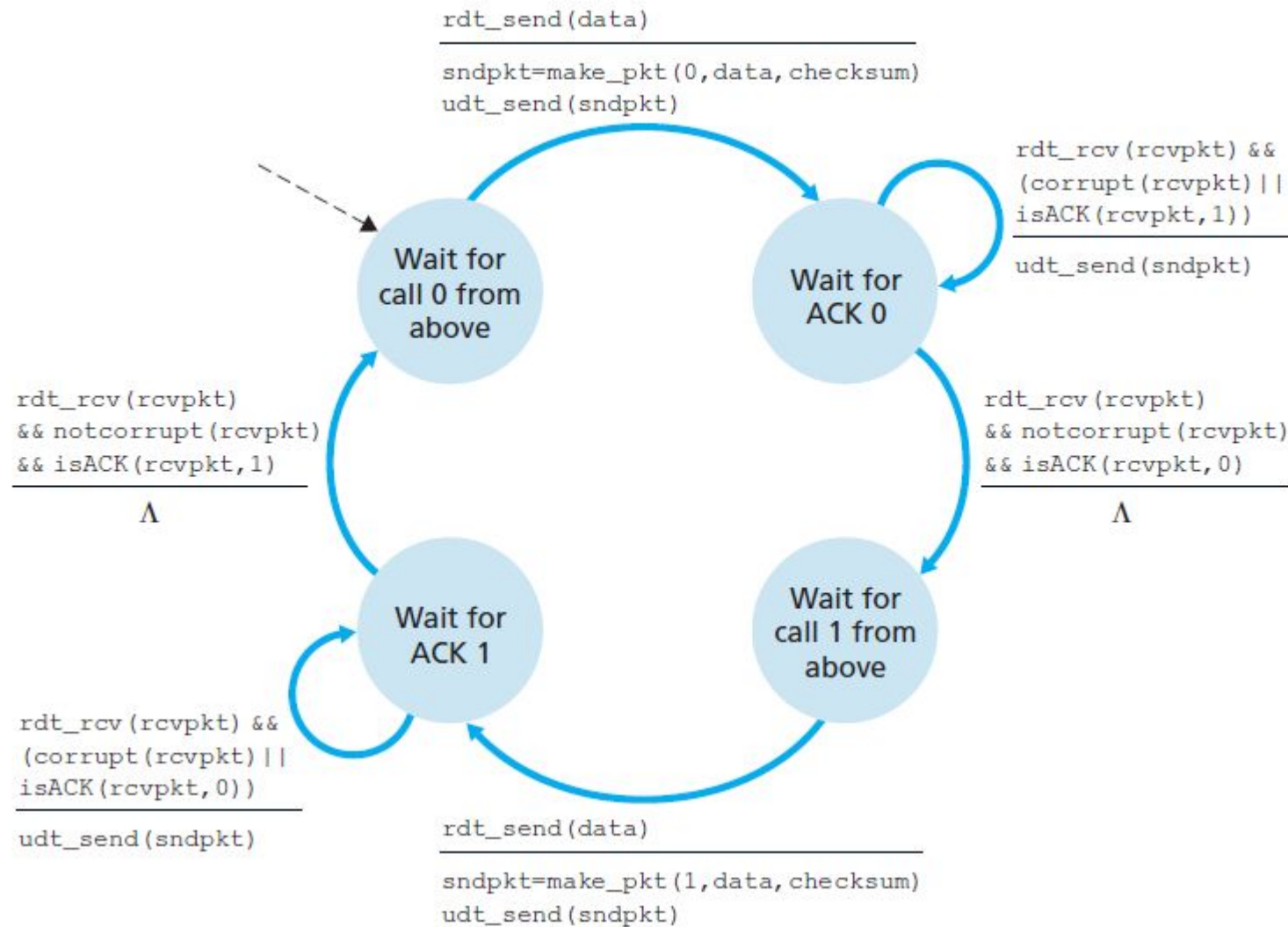
- seq # added to pkt
- two seq. #s (0,1) will suffice.
Why?
- must check if received ACK/NAK corrupted
- twice as many states
 - state must “remember” whether “expected” pkt should have seq # of 0 or 1

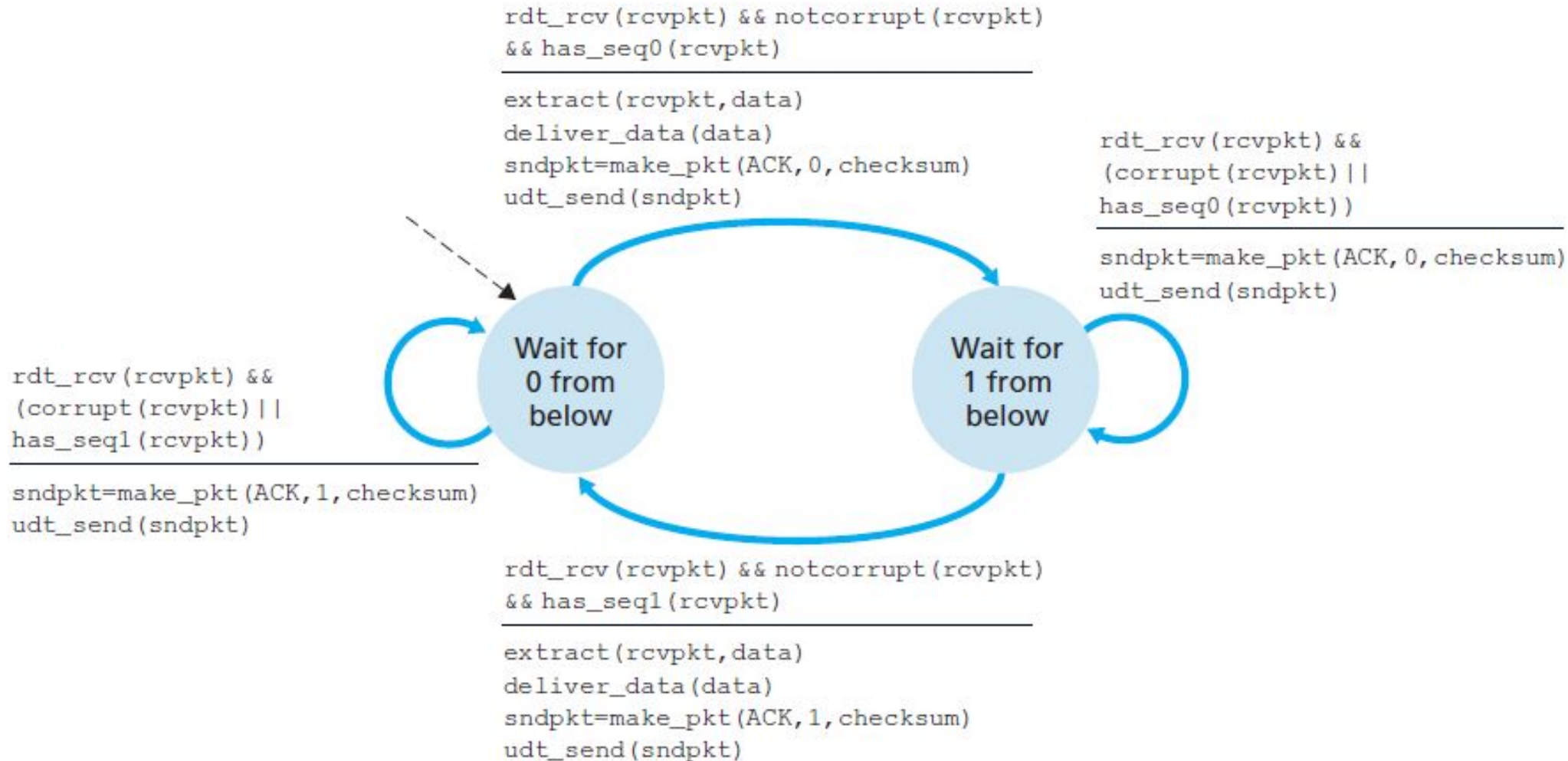
receiver:

- must check if received packet is duplicate
 - state indicates whether 0 or 1 is expected pkt seq #
- note: receiver can *not* know if its last ACK/NAK received OK at sender

- same functionality as rdt2.1, using ACKs only
- instead of NAK, receiver sends ACK for last pkt received OK
 - receiver must *explicitly* include seq # of pkt being ACKed
- duplicate ACK at sender results in same action as NAK:
retransmit current pkt

As we will see, TCP uses this approach to be NAK-free





New channel assumption: underlying channel can also lose packets (data, ACKs)

- checksum, sequence #s, ACKs, retransmissions will be of help ... but not quite enough

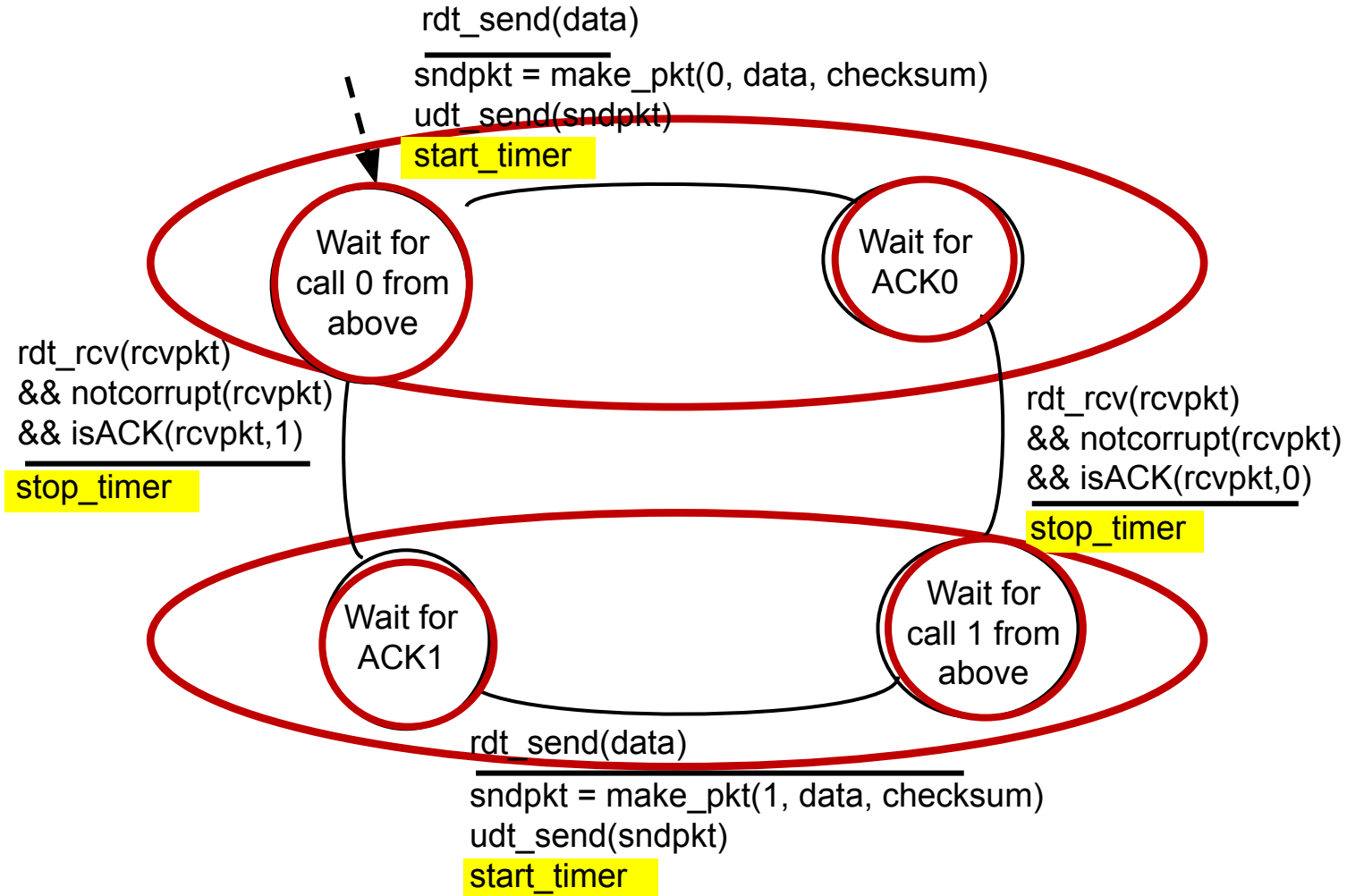
Q: How do *humans* handle lost sender-to-receiver words in conversation?

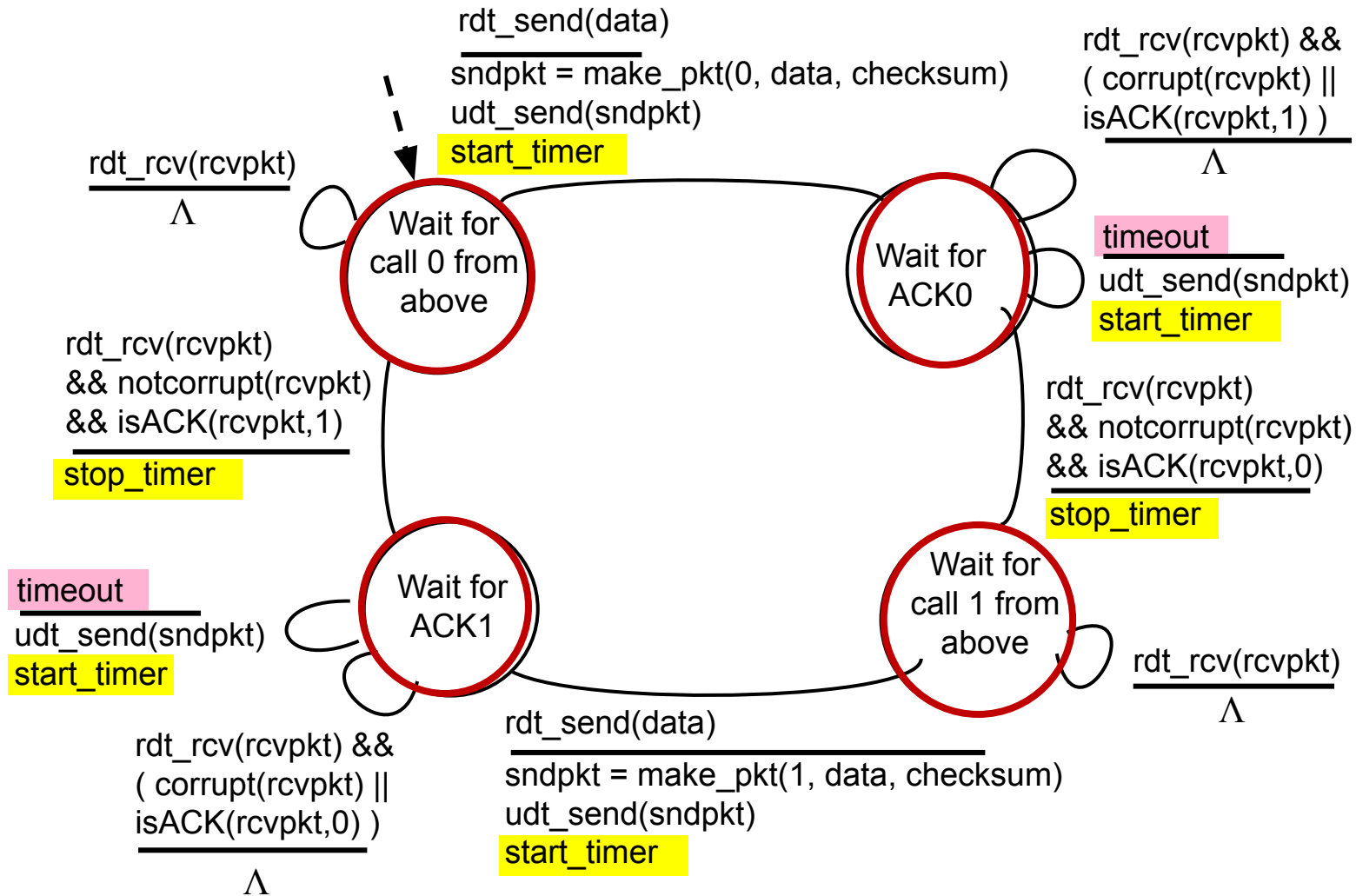
Approach: sender waits “reasonable” amount of time for ACK

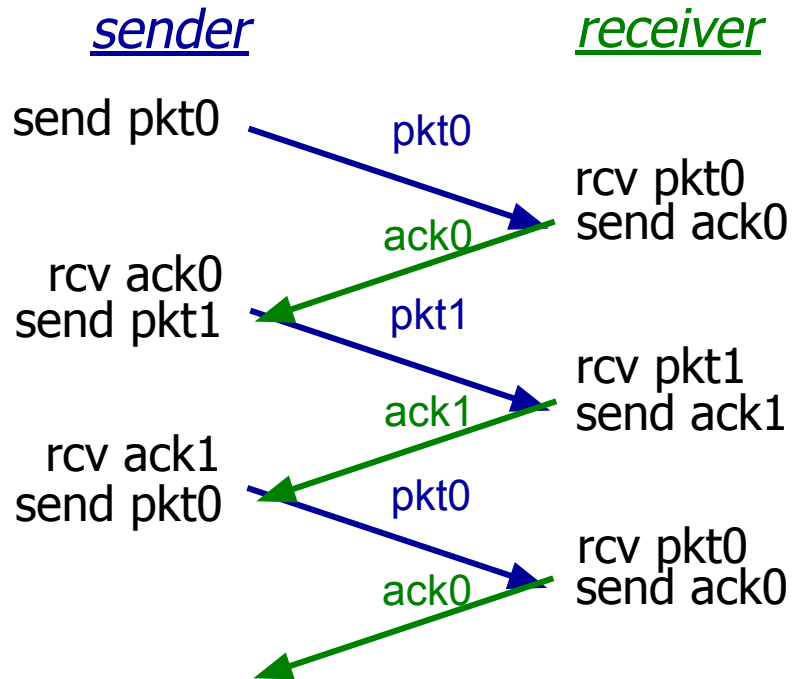
- retransmits if no ACK received in this time
- if pkt (or ACK) just delayed (not lost):
 - retransmission will be duplicate, but seq #s already handles this!
 - receiver must specify seq # of packet being ACKed
- use countdown timer to interrupt after “reasonable” amount of time



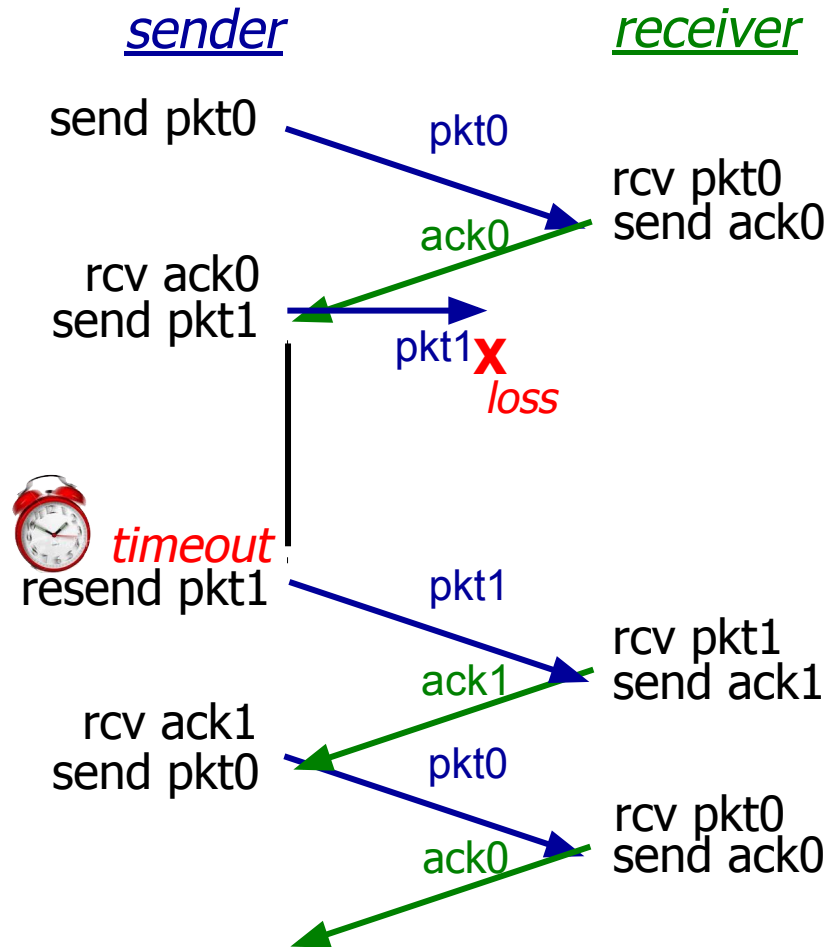
timeout



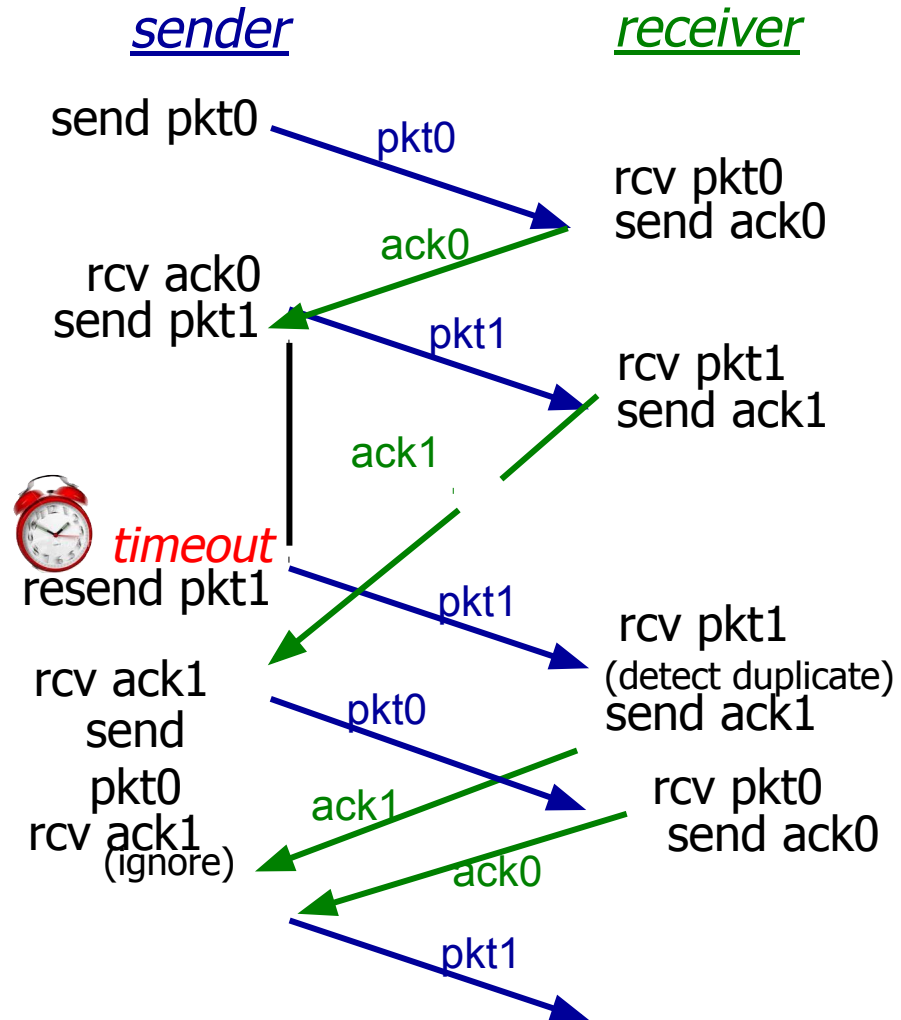
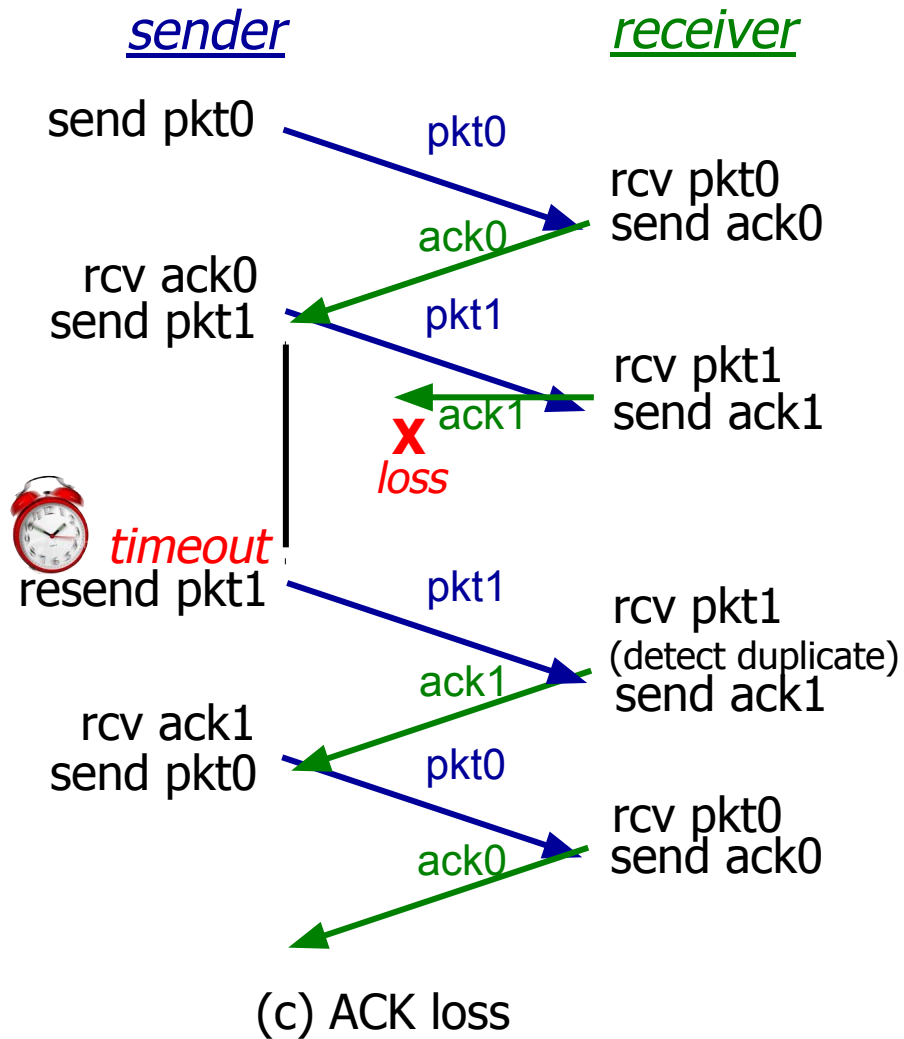




(a) no loss

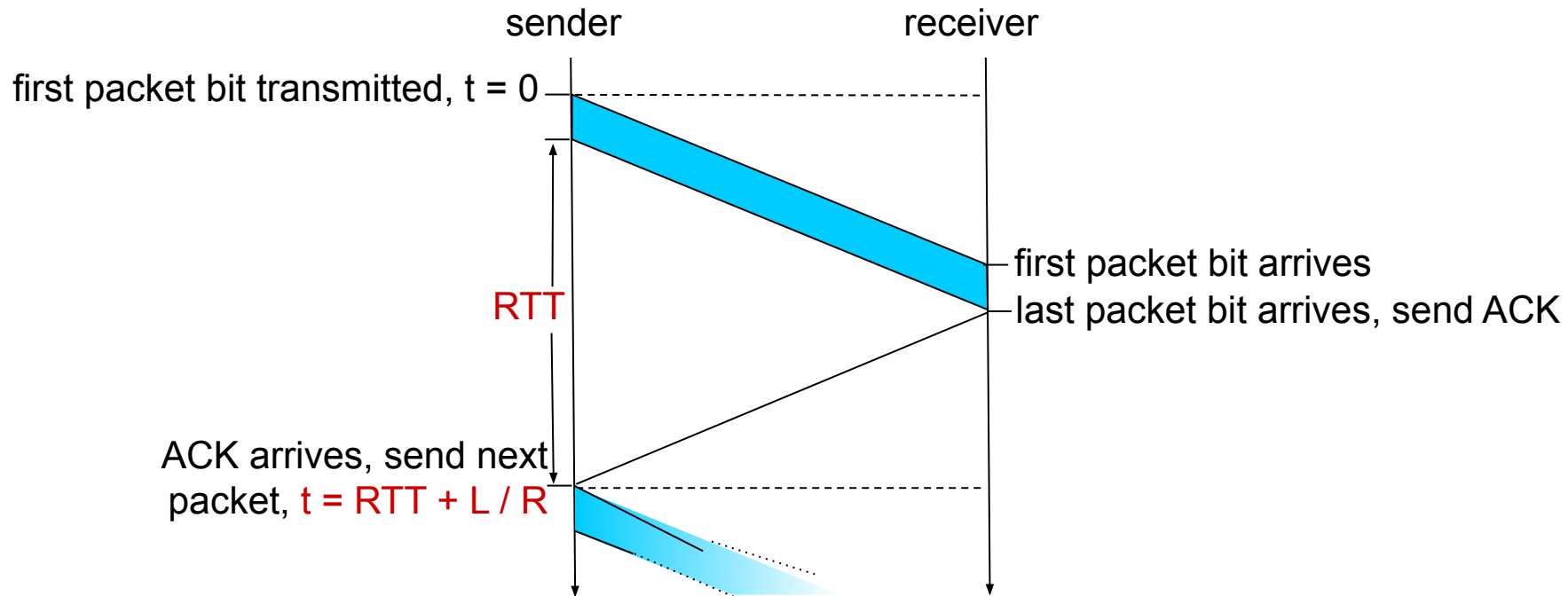


(b) packet loss

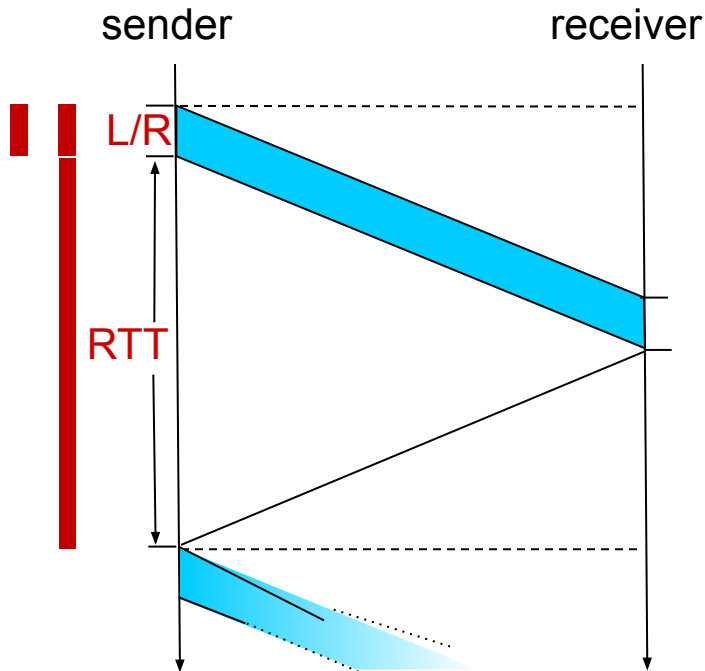


- U_{sender} : *utilization* – fraction of time sender busy sending
- example: 1 Gbps link, 15 ms prop. delay, 8000 bit packet
 - time to transmit packet into channel:

$$D_{trans} = \frac{L}{R} = \frac{8000 \text{ bits}}{10^9 \text{ bits/sec}} = 8 \text{ microsecs}$$



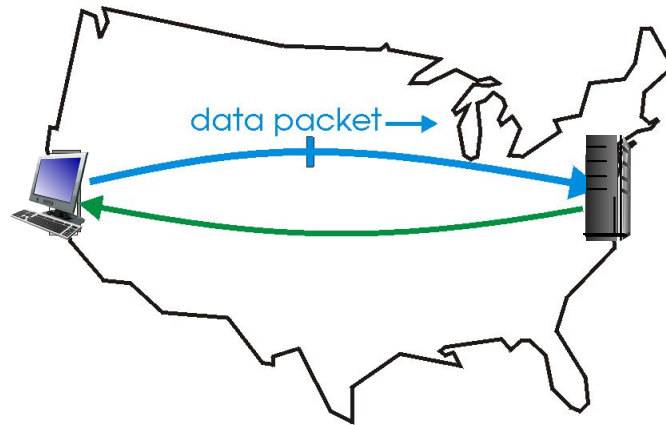
$$\begin{aligned}U_{\text{sender}} &= \frac{L / R}{RTT + L / R} \\&= \frac{.008}{30.008} \\&= 0.00027\end{aligned}$$



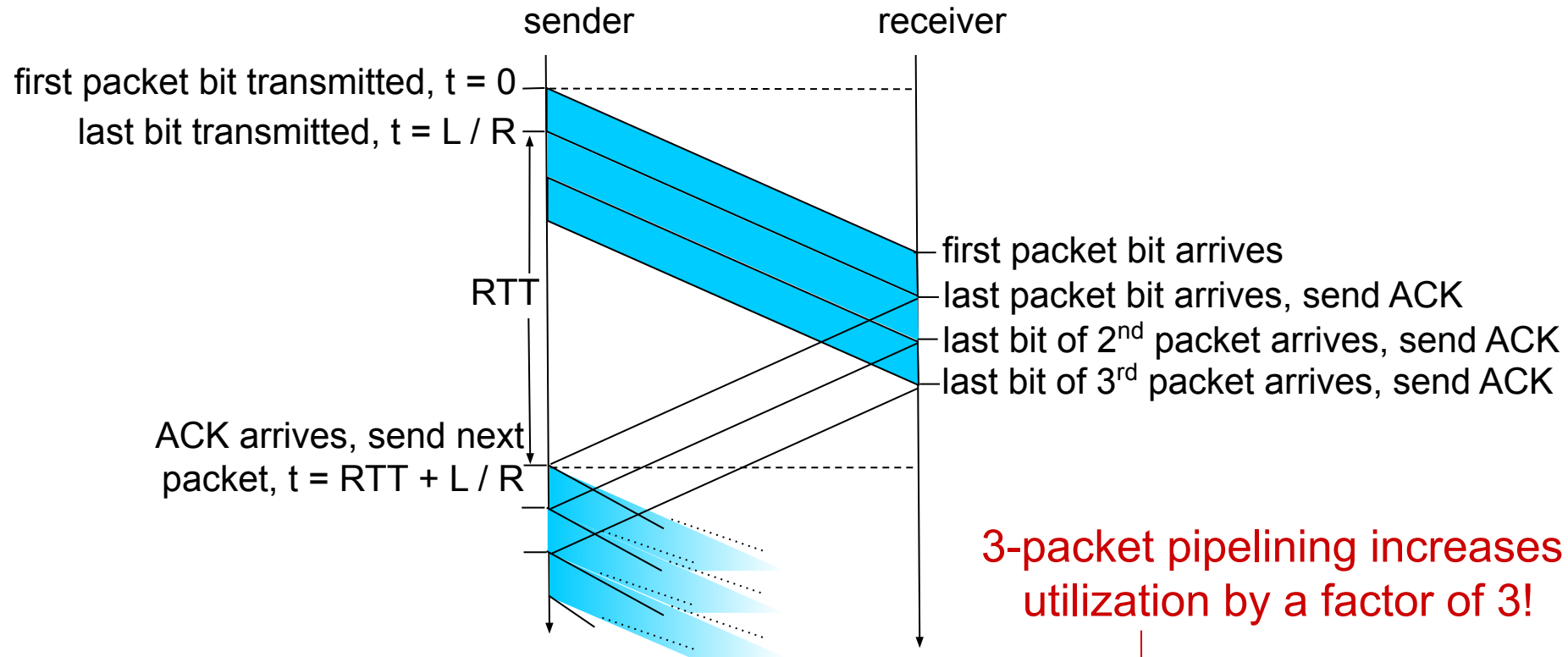
- rdt 3.0 protocol performance stinks!
- Protocol limits performance of underlying infrastructure (channel)

pipelining: sender allows multiple, “in-flight”, yet-to-be-acknowledged packets

- range of sequence numbers must be increased
- buffering at sender and/or receiver



(a) a stop-and-wait protocol in operation



3-packet pipelining increases utilization by a factor of 3!

$$U_{\text{sender}} = \frac{3L / R}{RTT + L / R} = \frac{.0024}{30.008} = 0.00081$$



THANK YOU

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